

EN 512: Structure-Property Correlations for Energy Applications

Course Code	EN 512
Course Name	Structure-Property Correlations for Energy Applications
L-T-P-C	1-0-4-3
Prerequisites	IC241 (Materials Science for Engineers)/ Instructor Consent
Intended for	UG/PG (Compulsory for MTech. in Materials and Energy Engineering, and Elective for others)
Mutual Exclusion	None .
Approval	9th Senate (May 2016)
Revised	44th BoA (Nov 2021) - revised

Course Contents

- **Module - 1:** Introduction to energy materials; photovoltaics, electrochemical systems, thermoelectrics, solar thermal systems. Importance of materials characterization (2 hours).
- **Module - 2:** Structure of Materials – Crystal structure, micro-structure, and macrostructure; Determination of crystal structure by diffraction, X ray diffraction and electron and neutron diffraction; Vibrational spectroscopy (IR and Raman spectroscopy) for structural characterization of materials. (4 Hours)
- **Module - 3:** Microstructure determination by light, and electron microscopy (SEM and TEM); binary alloys and distribution of phases in microstructure. (2 Hours)
- **Module - 4:** Thermal analysis by TGA-DSC; Determination of enthalpy, melting, decomposition and phase transition temperatures. (2 Hours)
- **Module - 5:** Mechanical property determination by indentation and tensile test, Stress-Strain diagram (2 Hours)
- **Module - 6:** Basics of electronic band structure, Electrical and optical properties of energy materials, determination of bandgap, Absorption (UV-Vis), emission (Photoluminescence) (2 Hours)

Laboratory/practical/tutorial Modules: [42 Hours]

- **Structural Characterization (Module-1):**
 - Preparation and processing of specimen and structural characterization using XRD, Determination of crystal structure by X-ray diffraction in a diffractometer
 - Preparation and processing of specimen and structural characterization using FT-IR, and Raman spectroscopy
- **Microstructural Characterization (Module-2):**
 - Preparation and processing of specimen and observation of microstructure in single phase alloy under optical microscope, scanning electron microscope
 - Preparation and processing of specimen and observation of microstructure and electron diffraction using transmission electron microscope
- **Thermal Characterization: (Module-3):**
 - Preparation and processing of specimen and thermal characterization using TGA-DSC
 - Thermal conductivity measurement
- **Mechanical Characterization (Module-4):**
 - Preparation and processing of specimen and characterization of mechanical property using tensile test UTM
 - preparation and processing of specimen and elastic modulus as well as hardness measurements by indentation method.
- **Electrical and Optical Characterization (Module 5):**
 - Preparation and processing of specimen and characterization of Current voltage measurement, electrical conductivity, opto-electrical property.
 - Preparation and processing of specimen for Absorption (UV-Vis), emission (Photoluminescence/fluorescence) spectroscopy and determination of band gap.

Text Books

1. Robert E Reed-Hill and Reza Abbaschian, **Physical Metallurgy Principles**, Thomson, 2003 reprint.
2. R. E. Hummel, **Electronic Properties of Materials**, Springer, 4th ed. 2011

References

1. Mauro Sardela, **Practical Materials Characterization**, Springer New York, 2014
2. A.R. West, **Solid-State Chemistry and Its Applications**, Wiley, 2014